

Usability Assessment of Knowledge-Based Systems in Customer Service Process Re-engineering

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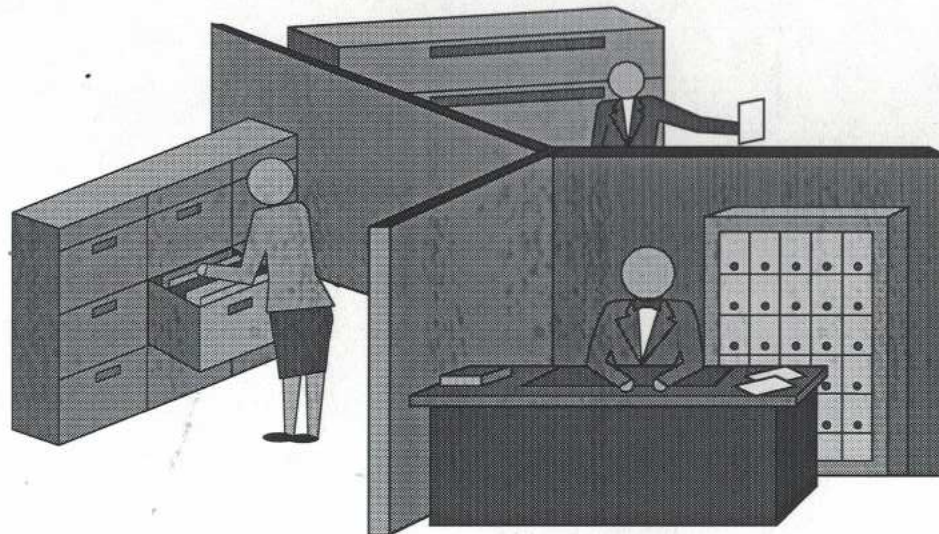
This seminar describes a set of ^{representative} ~~exemplary~~ techniques of usability engineering in the development of a knowledge-based system for consultative sales. The methods we use include user-centered and participatory design, early focus on design and assessment for usability and progressive prototyping.

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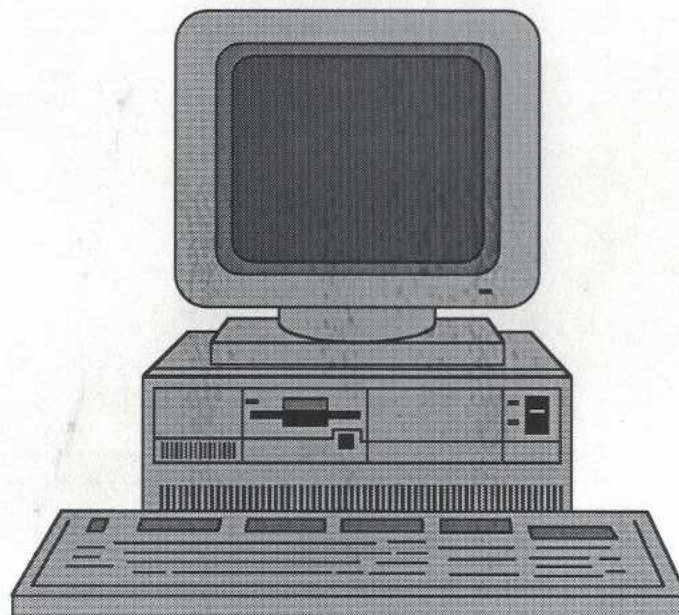
Customer Service Process Re-engineering

provide total customer service and *satisfaction* through the development of *effective* and *efficient* processes



Knowledge-based Systems (consultative sales)

- **capture, distribute and enhance consultative sales knowledge**
- **quality recommendation and advice to external customers**



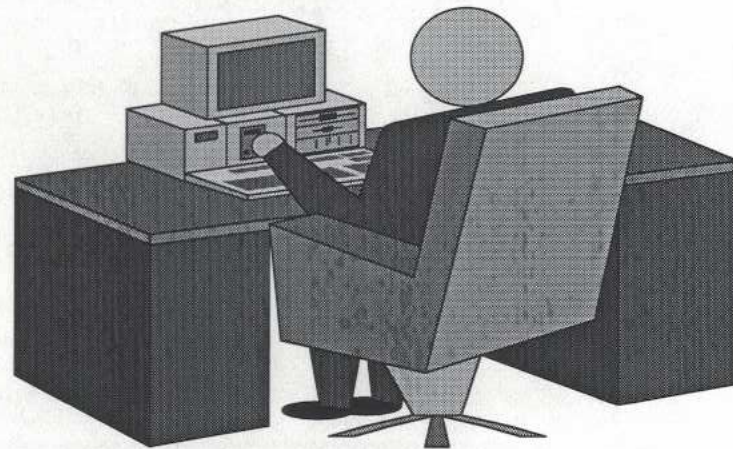
Usability Criteria

Measurements of Usability (ISO 9241)

- Effectiveness
- Efficiency
- Satisfaction

Players

- Users
- External Customers
- Developers
- Management



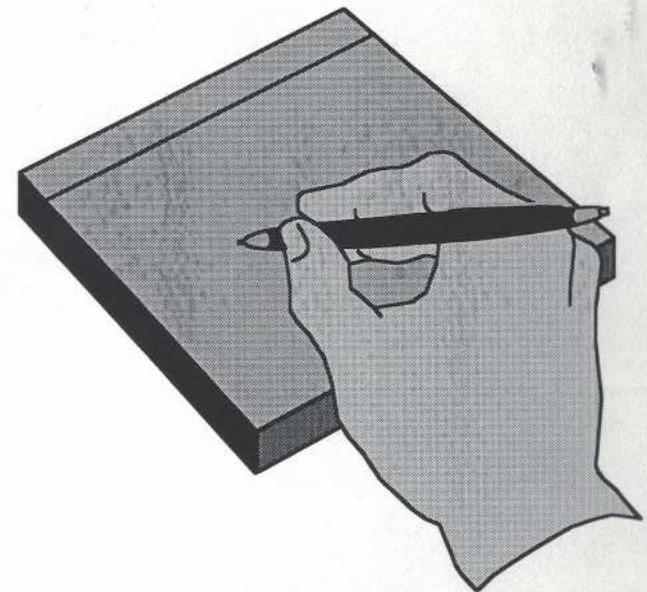
Usability Engineering Environment

- **User-Centered Approach & Participatory Design**
- **Progressive Prototyping with Iterative Design**
- **Early Focus on Design for Usability and Integrative Usability Assessment**



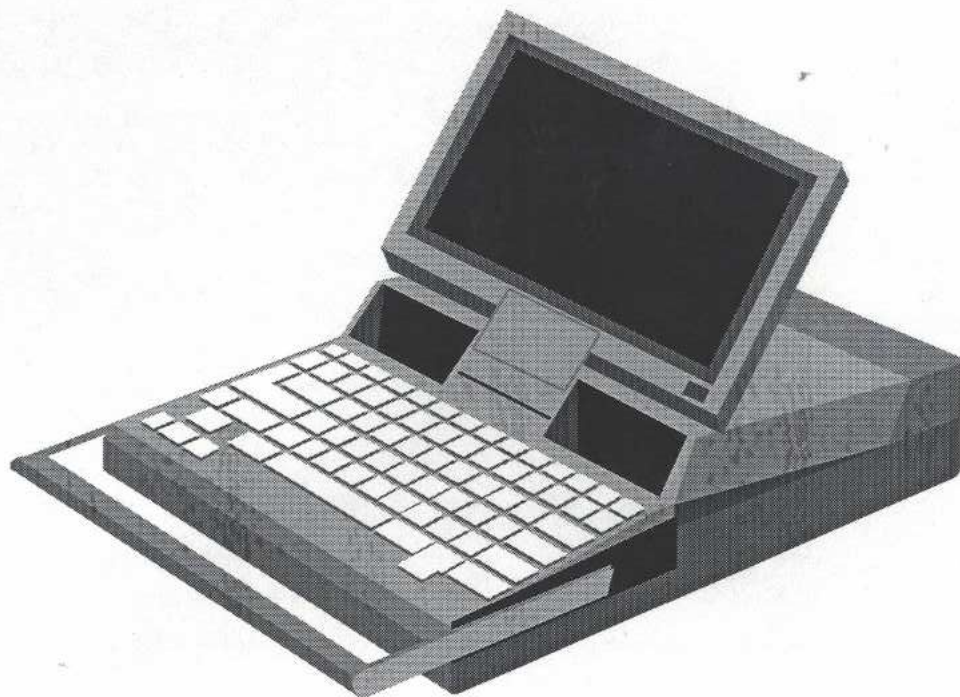
Usability Studies of Low-Fidelity Prototypes

- Paper and Pencil Scenario Simulation
- Work Flow Analysis
- Behavioral
 - Time to complete scenario
 - Errors in locating fields of data entry
- Attitudinal
 - Subjective usability rating
 - interviews



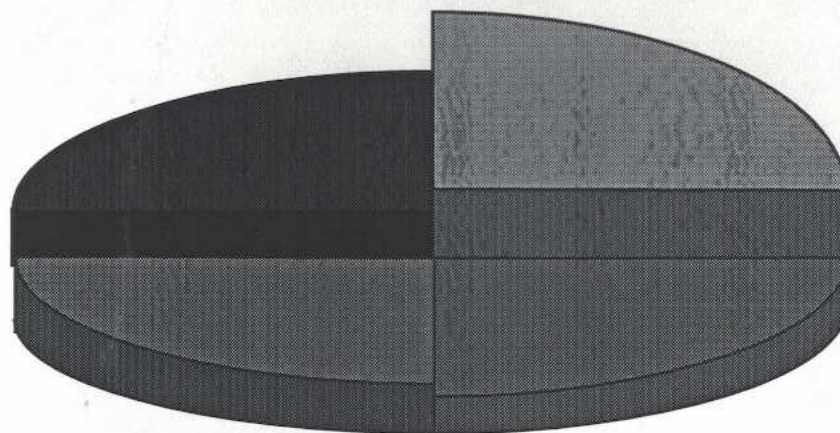
Usability Studies of High-Fidelity Prototypes

- Time Measurement
- Work Flow Analysis
- Naturalistic Observation
- Interviews



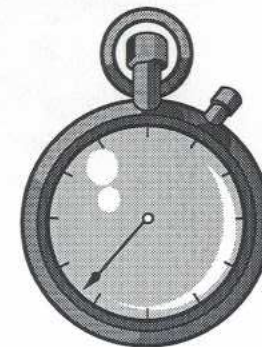
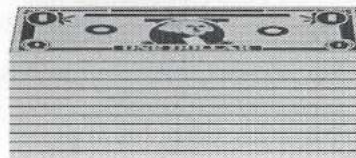
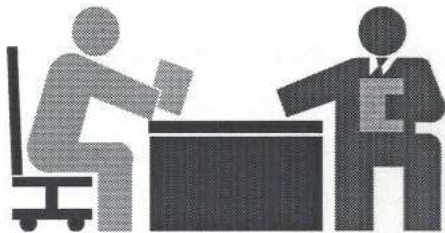
Baseline Measurement

- Validity of Solution Recommendation
- Usage of Knowledge-based system
- Customer Satisfaction
- Revenue and product sales



General Constraints and Limitations

- Confounding factors - changing market conditions, simultaneous introduction of other systems, changing corporate policies...
- Availability of users
- Time, resource and budget



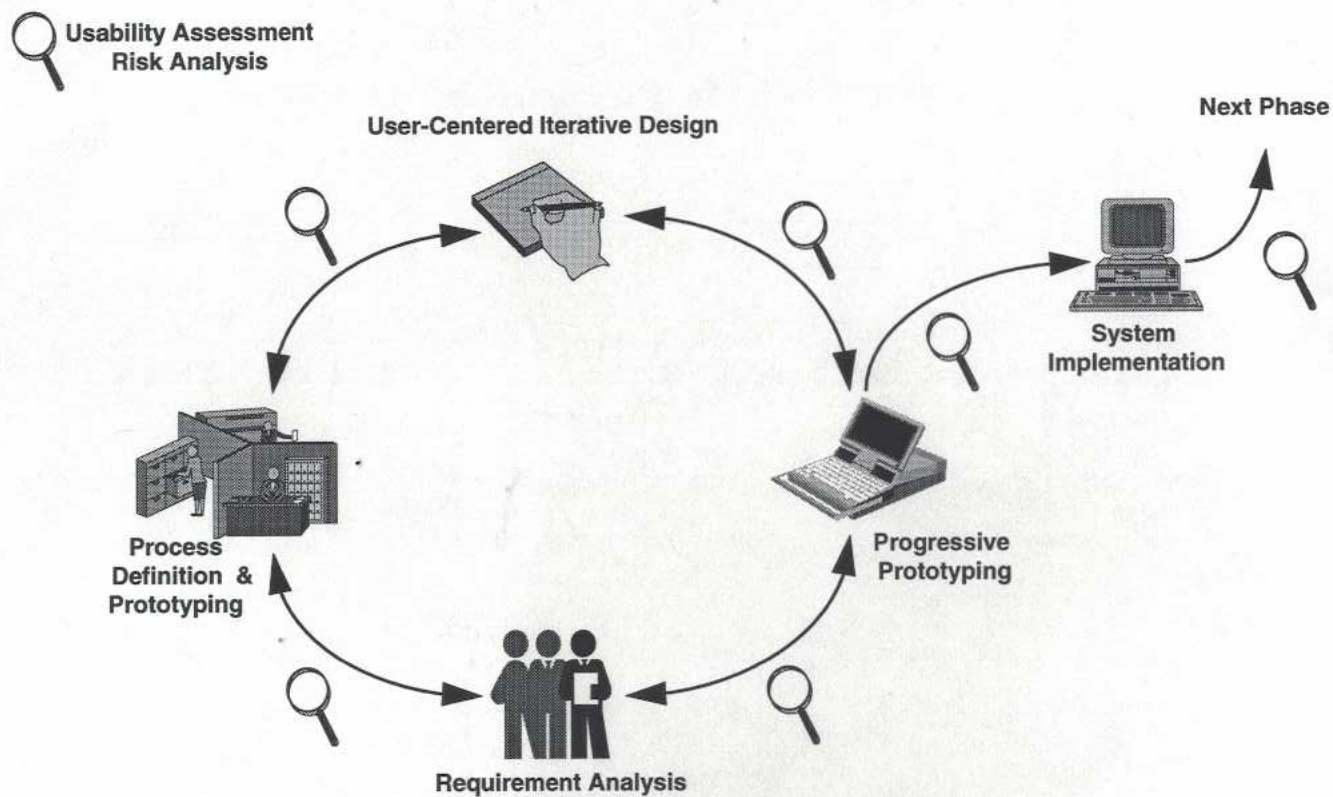


Figure 1. High-level Usability Engineering Cycle